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SYSTEMS OF LINEAR EQUATIONS

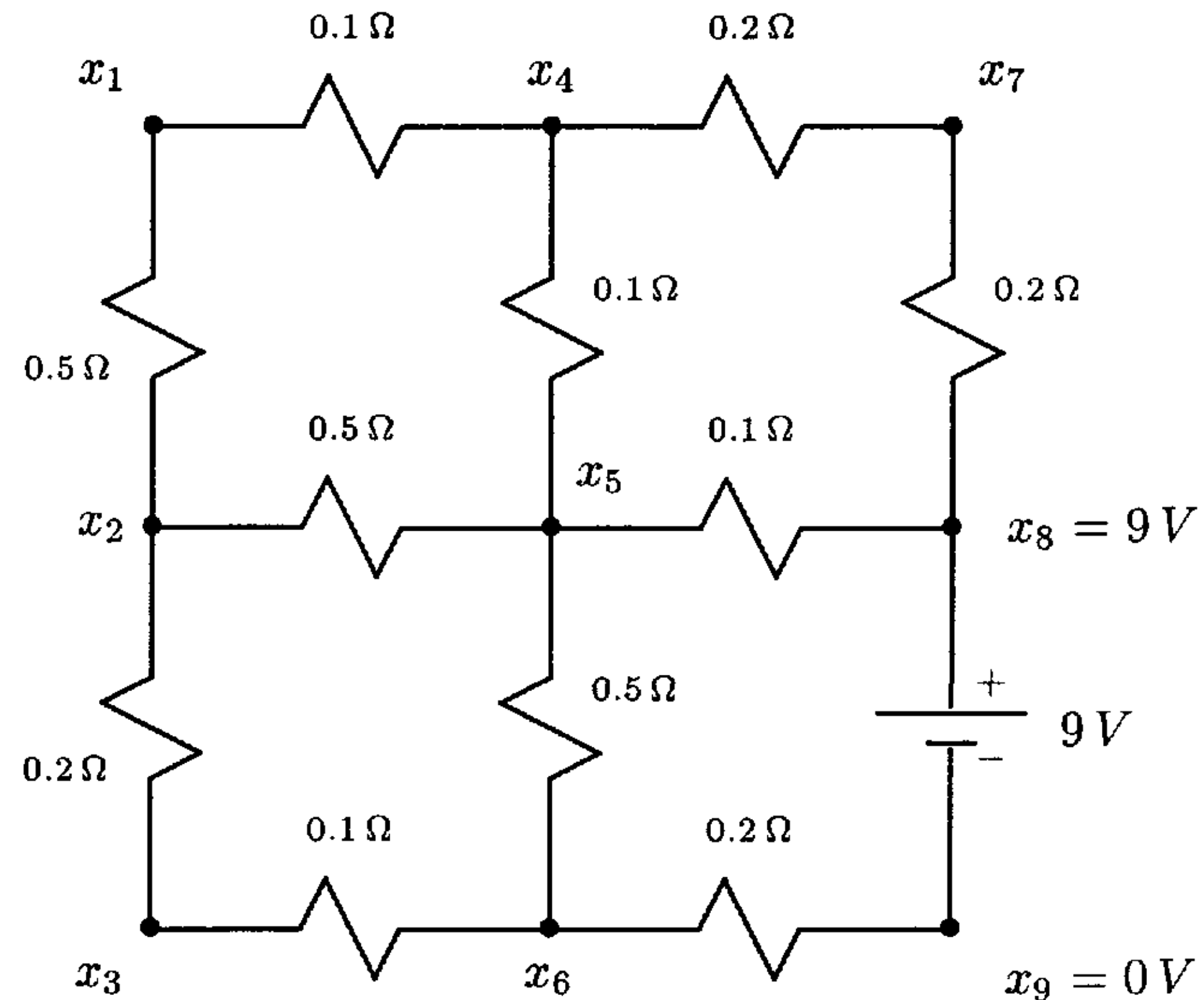
GAUSSIAN ELIMINATION AND ITS VARIANTS

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TUTORIAL

GAUSSIAN ELIMINATION AND ITS VARIANTS

Exercise 1 Consider the electrical circuit in the figure below



Electric circuit with nodal voltages

- Write down a linear system $\mathbf{Ax} = \mathbf{b}$ with seven equations for the seven unknowns nodal voltages.
- Using Julia, for solving the system to find the nodal voltages, calculate the residual $\mathbf{r} = \mathbf{b} - \mathbf{A}\hat{\mathbf{x}}$, where $\hat{\mathbf{x}}$ denote your computed solution.
- In theory $\mathbf{r}$ should be zero. In practice you will get a thing but nonzero residual because of the roundoff errors in your computations.

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Exercise 2: write a Julia program for forward substitution. Use pencil and paper to solve the system

Estimate the execution of the forward substitution by counting the floating-point operations (flops)

$$\begin{bmatrix} 2 & 0 & 0 & 0 \\ -1 & 2 & 0 & 0 \\ 3 & 1 & -1 & 0 \\ 4 & 1 & -3 & 3 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 2 \\ 9 \end{bmatrix}$$

Exercise 3

```
for  $i = 1, \dots, n$   
  for  $j = 1, \dots, i - 1$  (not executed when  $i = 1$ )  
     $b_i \leftarrow b_i - g_{ij}b_j$   
  if  $g_{ii} = 0$ , set error flag, exit  
   $b_i \leftarrow b_i / g_{ii}$ 
```

Write a modified version of the algorithm on the right that checks for leading zeros in b and take appropriate action.

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Exercise 4

1. Develop the row-oriented version of the back substitution process. Write pseudocode of it and a Julia program.
2. Develop the column-oriented version of the back substitution process. Write pseudocode of it and a Julia program.
3. Using your Julia codes in 1) and 2) solve the upper-triangular system

$$\begin{bmatrix} 3 & 2 & 1 & 0 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & -2 & 1 \\ 0 & 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -10 \\ 10 \\ 1 \\ 12 \end{bmatrix}$$

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Exercise 5 Implement the below algorithm in Julia and use your program to solve the system

$$\mathbf{Ax} = \mathbf{b}.$$

Cholesky's Algorithm (inner-product form)

for $i = 1, \dots, n$
 for $k = 1, \dots, i - 1$ (not executed when $i = 1$)
 $a_{ii} \leftarrow a_{ii} - a_{ki}^2$
 if $a_{ii} \leq 0$, set error flag (A is not positive definite), exit
 $a_{ii} \leftarrow \sqrt{a_{ii}}$ (this is r_{ii})
 for $j = i + 1, \dots, n$ (not executed when $i = n$)
 for $k = 1, \dots, i - 1$ (not executed when $i = 1$)
 $a_{ij} \leftarrow a_{ij} - a_{ki}a_{kj}$
 $a_{ij} \leftarrow a_{ij}/a_{ii}$ (this is r_{ij})

Where

$$A = \begin{bmatrix} 4 & -2 & 4 & 2 \\ -2 & 10 & -2 & -7 \\ 4 & -2 & 8 & 4 \\ 2 & -7 & 4 & 7 \end{bmatrix} \quad \text{and} \quad b = \begin{bmatrix} 8 \\ 2 \\ 16 \\ 6 \end{bmatrix}$$

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EXERCICE 6

Let

$$A = \begin{bmatrix} 16 & 4 & 8 & 4 \\ 4 & 10 & 8 & 4 \\ 8 & 8 & 12 & 10 \\ 4 & 4 & 10 & 12 \end{bmatrix} \quad \text{and} \quad b = \begin{bmatrix} 32 \\ 26 \\ 38 \\ 30 \end{bmatrix}.$$

Notice that A is symmetric. (a) Use the inner-product formulation of Cholesky's method to show that A is positive definite and compute its Cholesky factor. (b) Use forward and back substitution to solve the linear system $Ax = b$. $\square$

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Exercise 7

$$A = \begin{bmatrix} 2 & 1 & -1 & 3 \\ -2 & 0 & 0 & 0 \\ 4 & 1 & -2 & 6 \\ -6 & -1 & 2 & -3 \end{bmatrix} \quad \text{and} \quad b = \begin{bmatrix} 13 \\ -2 \\ 24 \\ -14 \end{bmatrix}.$$

- (a) Calculate the appropriate (four) determinants to show that A can be transformed to (nonsingular) upper-triangular form by operations of type 1 only. (By the way, this is strictly an academic exercise. In practice one never calculates these determinants in advance.)
- (b) Carry out the row operations of type 1 to transform the system $Ax = b$ to an equivalent system $Ux = y$, where U is upper triangular. Save the multipliers ~~in a separate list~~.
- (c) Carry out the back substitution on the system $Ux = y$ to obtain the solution of $Ax = b$. Don't forget to check your work.

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Exercise 8

$$A = \begin{bmatrix} 4 & -2 & 4 & 2 \\ -2 & 10 & -2 & -7 \\ 4 & -2 & 8 & 4 \\ 2 & -7 & 4 & 7 \end{bmatrix} \quad \text{and} \quad b = \begin{bmatrix} 8 \\ 2 \\ 16 \\ 6 \end{bmatrix}.$$

Use Cholesky's Algorithm to show that A is positive definite and use the Cholesky's factor to solve the system $Ax = b$

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Exercise 9

Use Gaussian elimination (by hand) with row interchange to that the matrix

$$A = \begin{bmatrix} 2 & 10 & 8 & 8 & 6 \\ 1 & 4 & -2 & 4 & -1 \\ 0 & 2 & 3 & 2 & 1 \\ 3 & 8 & 3 & 10 & 9 \\ 1 & 4 & 1 & 2 & 1 \end{bmatrix}$$

Is non singular. Find L and D and solve the system $Ax = b$, where

$$b = \begin{bmatrix} 52 \\ 14 \\ 12 \\ 51 \\ 15 \end{bmatrix}$$

Check you answer with Julia

Exercise 10 Consider the following data.

t_i	1.0	1.5	2.0	2.5	3.0
y_i	1.1	1.2	1.3	1.3	1.4

- (a) Set up an overdetermined system `for the data above` for a straight line passing through the data points. Use the standard basis polynomials $\phi_1(t) = 1$, $\phi_2(t) = t$.
- (b) Use `Julia` to calculate the least-squares solution of the system from part (a). This is a simple matter. Given an overdetermined system $Ax = b$, the `Julia` command `x = A\b` computes the least squares solution. Recall that this is exactly the same command as would be used to tell `Julia` to solve a square system $Ax = b$ by Gaussian elimination.¹⁶ Some useful `Julia` commands:

```
t = 1:.5:3; t = t'; s = ones(5,1); A = [s t];
```


Exercise 10 continued

We already know that `Julia` uses Gaussian elimination with partial pivoting in the square case. In the next two sections you will find out what `Julia` does in the overdetermined case.

- (c) Use the `Julia` `plot` command to plot the five data points and your least squares straight line. Type `? plot` for information about using the `plot` command, or search in `Julia`'s help browser.
- (d) Use `Julia` to compute $\|r\|_2$, the norm of the residual.

Exercise 11 Suppose $\hat{A}$ is obtained from A by interchanging the i th and j th rows. This is an operation of type 2.

- (a) Show that $\hat{A} = MA$, where M is the matrix obtained from the identity matrix by interchanging the i th and j th rows. We call this M an *elementary matrix* of type 2.
- (b) Show that $\det(M) = -1$ and $\det(\hat{A}) = -\det(A)$. Thus we see (again) that $\hat{A}$ is nonsingular if and only if A is.
- (c) Show that $M^{-1} = M$. Explain why this result would have been expected, in view of the action of M as the effector of an elementary operation.

Exercise 12 Suppose $\hat{A}$ is obtained from A by multiplying the i th row by the nonzero constant c .

- (a) Find the form of the matrix M (an *elementary matrix* of type 3) such that $\hat{A} = MA$.
- (b) Find M^{-1} and state its function as an elementary matrix.
- (c) Find $\det(M)$ and determine the relationship between $\det(\hat{A})$ and $\det(A)$. Deduce that $\hat{A}$ is nonsingular if and only if A is.

Exercise 13 We have seen that if the leading principal submatrices A_k are nonsingular for $k = 1, \dots, n$, then A has an LU decomposition. Prove the following converse. If A is nonsingular and has an LU decomposition, then A_k is nonsingular for $k = 1, \dots, n$. (*Hint: Partition the matrices in the decomposition $A = LU$.*) $\square$

Exercise 14 Write an algorithm that implements Gaussian elimination with partial pivoting. Store L and U over A , and save a record of the row interchanges. $\square$

Your solution to Exercise 1.8.12 should look something like this:

```

Gauss ( $A, n, \text{intch}, \text{flag}$ )
clear  $\text{flag}$ 
for  $k = 1, 2, \dots, n - 1$ 
    [
         $\text{amax} \leftarrow \max\{|a_{kk}|, |a_{k+1,k}|, \dots, |a_{nk}|\}$ 
        if ( $\text{amax} = 0$ ) then
            [
                set  $\text{flag}$ 
                 $\text{intch}(k) \leftarrow 0$ 
            ]      ( $A$  is singular)
        else
            [
                 $m \leftarrow$  smallest integer  $\geq k$  for which  $|a_{mk}| = \text{amax}$ 
                 $\text{intch}(k) \leftarrow m$       (record row interchange)
                if ( $m \neq k$ ) then
                    [ interchange rows  $k$  and  $m$ , including multipliers
                    for  $i = k + 1, \dots, n$ 
                        [  $a_{ik} \leftarrow a_{ik}/a_{kk}$       calculate multipliers
                        for  $i = k + 1, \dots, n$ 
                            [ subtract  $a_{ik}$  times  $[a_{k,k+1}, \dots, a_{k,n}]$ 
                                from  $[a_{i,k+1}, \dots, a_{i,n}]$ 
                            ]
                        ]
                    ]      (row operations)
            ]
        if ( $a_{nn} = 0$ ) then
            [
                set  $\text{flag}$ 
                 $\text{intch}(n) \leftarrow 0$ 
            ]      ( $A$  is singular)
        else
            [  $\text{intch}(n) \leftarrow n$  ]
    ]

```

Write a Julia program, that implement this algorithm

If A is found to be singular, the algorithm sets a flag but does not stop. It finishes the LU decomposition, but U is singular. The array A now contains the LU decomposition of $\hat{A} = PA$, and the array *intch* contains a record of the row interchanges (i.e. an encoding of the permutation matrix P).